DAY 4 - Linux Commands

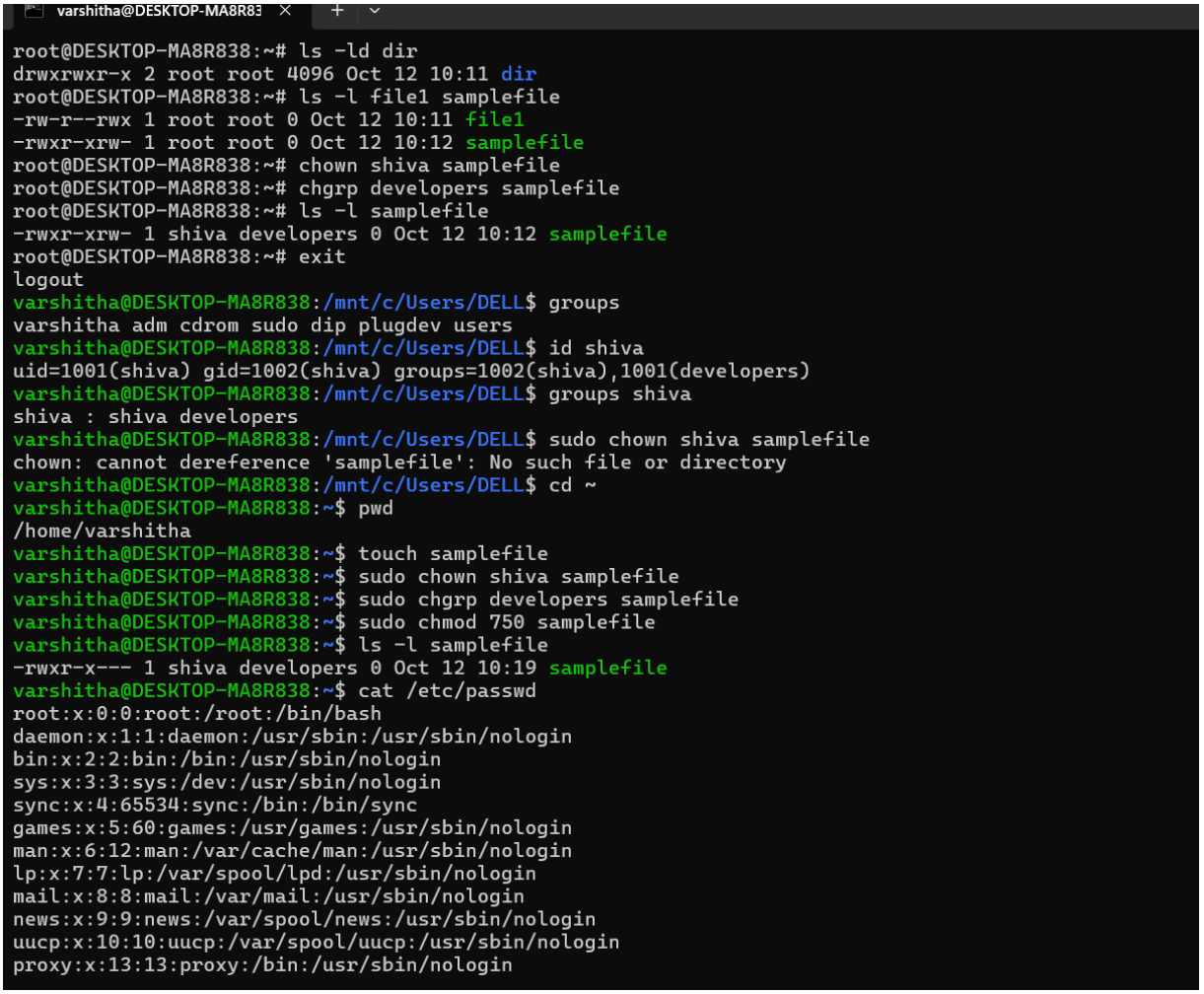
```
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ sudo useradd -m shiva
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ sudo passwd shiva
New password:
Retype new password:
passwd: password updated successfully
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ sudo usermod -aG devops shiva
usermod: group 'devops' does not exist
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ sudo usermod -aG developers shiva
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ groups shiva
shiva : shiva developers
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ id shiva
uid=1001(shiva) gid=1002(shiva) groups=1002(shiva),1001(developers)
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ sudo su -
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 6.18.40.1-microsoft-standard-WSL2 x86_64)

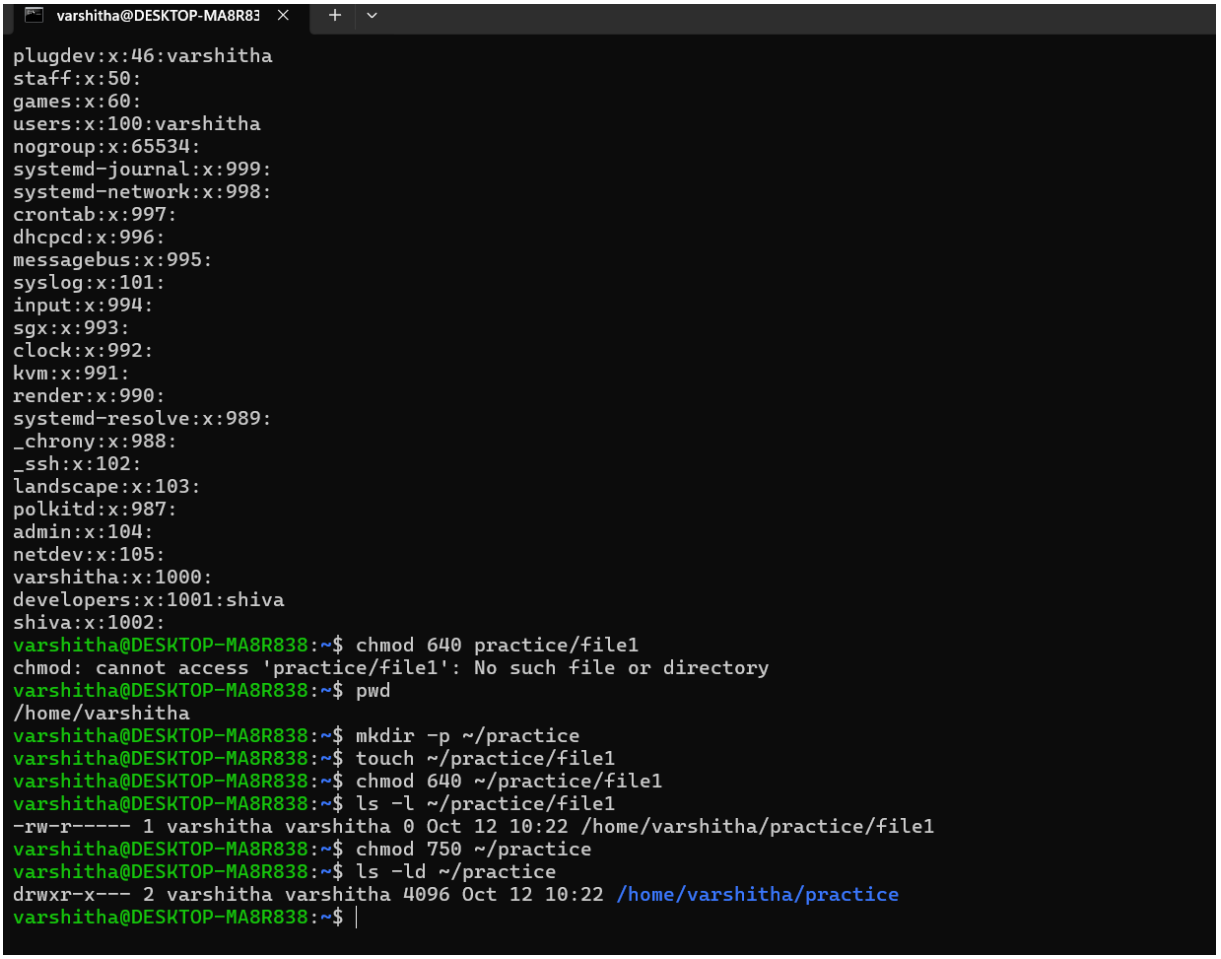
 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Oct 12 10:10:19 UTC 2026

  System load:  0.08               Processes:             36
  Usage of /:   0.1% of 1006.85GB  Users logged in:       0
  Memory usage: 9%                 IPv4 address for eth0: 192.168.15.252
  Swap usage:   0%


This message is shown once a day. To disable it please create the
/root/.hushlogin file.
root@DESKTOP-MA8R838:~# sudo -u shiva whoami
shiva
root@DESKTOP-MA8R838:~# chmod g+w dir
chmod: cannot access 'dir': No such file or directory
root@DESKTOP-MA8R838:~# mkdir dir
root@DESKTOP-MA8R838:~# touch file1
root@DESKTOP-MA8R838:~# touch samplefile
root@DESKTOP-MA8R838:~# chmod g+w dir
root@DESKTOP-MA8R838:~# chmod o+wx file1
root@DESKTOP-MA8R838:~# chmod u=rwx,g=rx,o=rw samplefile
root@DESKTOP-MA8R838:~# ls -ld dir
drwxrwxr-x 2 root root 4096 Oct 12 10:11 dir
```


varshitha@DESKTOP-MA8R83
root@DESKTOP-MA8R838:~# ls -ld dir
drwxrwxr-x 2 root root 4096 Oct 12 10:11 dir
root@DESKTOP-MA8R838:~# ls -l file1 samplefile
-rw-r--rwx 1 root root 0 Oct 12 10:11 file1
-rwxr-xrw- 1 root root 0 Oct 12 10:12 samplefile
root@DESKTOP-MA8R838:~# chown shiva samplefile
root@DESKTOP-MA8R838:~# chgrp developers samplefile
root@DESKTOP-MA8R838:~# ls -l samplefile
-rwxr-xrw- 1 shiva developers 0 Oct 12 10:12 samplefile
root@DESKTOP-MA8R838:~# exit
logout
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ groups
varshitha adm cdrom sudo dip plugdev users
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ id shiva
uid=1001(shiva) gid=1002(shiva) groups=1002(shiva),1001(developers)
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ groups shiva
shiva : shiva developers
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ sudo chown shiva samplefile
chown: cannot dereference 'samplefile': No such file or directory
varshitha@DESKTOP-MA8R838:/mnt/c/Users/DELL$ cd ~
varshitha@DESKTOP-MA8R838:~$ pwd
/home/varshitha
varshitha@DESKTOP-MA8R838:~$ touch samplefile
varshitha@DESKTOP-MA8R838:~$ sudo chown shiva samplefile
varshitha@DESKTOP-MA8R838:~$ sudo chgrp developers samplefile
varshitha@DESKTOP-MA8R838:~$ sudo chmod 750 samplefile
varshitha@DESKTOP-MA8R838:~$ ls -l samplefile
-rwxr-x--- 1 shiva developers 0 Oct 12 10:19 samplefile
varshitha@DESKTOP-MA8R838:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin



Create and Modify a Custom Docker Image by Installing and Removing Packages

```
ubuntu@ip-172-31-12-203:~/docker-package-lab$ sudo docker images

IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:8.0          7dcddc01f13b       1.1GB          249MB    U
package-lab:v1     a4a6e7c80d4f        313MB         76.9MB    U
ubuntu:24.04       534baea6a22c        119MB         31.7MB
wordpress:latest   81a34254acaa       1.12GB          284MB    U
ubuntu@ip-172-31-12-203:~/docker-package-lab$ sudo docker run -it --name package-container-v2 package-lab:v2
Unable to find image 'package-lab:v2' locally
docker: Error response from daemon: pull access denied for package-lab, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
ubuntu@ip-172-31-12-203:~/docker-package-lab$ ^C
ubuntu@ip-172-31-12-203:~/docker-package-lab$ sudo docker images

IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:8.0          7dcddc01f13b       1.1GB          249MB    U
package-lab:v1     a4a6e7c80d4f        313MB         76.9MB    U
ubuntu:24.04       534baea6a22c        119MB         31.7MB
wordpress:latest   81a34254acaa       1.12GB          284MB    U
ubuntu@ip-172-31-12-203:~/docker-package-lab$ ^C
ubuntu@ip-172-31-12-203:~/docker-package-lab$ ^C
ubuntu@ip-172-31-12-203:~/docker-package-lab$ cat Dockerfile
```

```
ubuntu@ip-172-31-12-203:~$ sudo docker pull wordpress:latest
latest: Pulling from library/wordpress
df60b2c773ce: Pulling fs layer
0f209d1853df: Download complete
ecc510c1e359: Downloading  18.87MB/29.84MB
df60b2c773ce: Pull complete
4633f536ccde: Pull complete
928ddadd1b1e: Pull complete
4f4fb700ef54: Pull complete
f5da053af527: Pull complete
04b05f2ec306: Pull complete
439f2909ad34: Pull complete
0822cfc31c02: Pull complete
171eb0bc859e: Pull complete
50427e99fb9c: Pull complete
17aef3cdc1c9: Pull complete
3b4c83254f6c: Pull complete
292513814dbc: Pull complete
53a3637988ce: Pull complete
7a19dc4d0e05: Pull complete
10ab328c1454: Pull complete
c957944e4676: Pull complete
7a3652c884f1: Pull complete
cdee7d186a1a: Pull complete
5bd2aab6bd16: Pull complete
44256314083a: Pull complete
e209c2ae9ae6: Download complete
338133eaa18f: Download complete
Digest: sha256:81a34254acaa6ff3b9a67268c2745e9a041c4094cbe5ccd35b75ed3bbdc033b5
Status: Downloaded newer image for wordpress:latest
docker.io/library/wordpress:latest
ubuntu@ip-172-31-12-203:~$ sudo docker pull mysql:8.0
8.0: Pulling from library/mysql
6ef6c7b50a93: Pull complete
96d30d9fbee8: Pull complete
e3e5d1ac74c1: Pull complete
297d04cfe470: Pull complete
4c8a3e0d4e4b: Pull complete
a63160a5eda1: Pull complete
7534d1db9f8d: Pull complete
49ec2dab01d9: Pull complete
0d74d296605b: Pull complete
ab24264a27e9: Pull complete
edf85873f64e: Pull complete
796812c73292: Download complete
af166387641d: Download complete
Digest: sha256:7dcddc01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
```

```
root@288d410afe39:/# vim --version
wget --version
git --version
php -v
VIM - Vi IMproved 9.1 (2024 Jan 02, compiled Aug 24 2026 22:13:04)
Included patches: 1-16, 647, 678, 697
Modified by team+vim@tracker.debian.org
Compiled by team+vim@tracker.debian.org
Huge version without GUI.  Features included (+) or not (-):
+acl               +file_in_path      +mouse_urxvt       -tag_any_white
+arabic            +find_in_path      +mouse_xterm       -tcl
+autocmd           +float             +multi_byte        +termguicolors
+autochdir         +folding           +multi_lang        +terminal
-autoservername    -footer            -mzscheme          +terminfo
-balloon_eval      +fork()            +netbeans_intg     +termresponse
+balloon_eval_term +gettext           +num64             +textobjects
-browse            -hangul_input      +packages          +textprop
++builtin_terms    +iconv             +path_extra        +timers
+byte_offset       +insert_expand     -perl              +title
+channel           +ipv6              +persistent_undo   -toolbar
+cindent           +job               +popupwin          +user_commands
-clientserver      +jumplist          +postscript        +vartabs
-clipboard         +keymap            +printer           +vertsplit
+cmdline_compl     +lambda            +profile           +vim9script
+cmdline_hist      +langmap           -python            +viminfo
+cmdline_info      +libcall           +python3           +virtualedit
```

```
ubuntu@ip-172-31-12-203: ~/docker-package-lab
  GNU nano 7.2                                  Dockerfile

FROM ubuntu:24.04

RUN apt-get update && apt-get install -y \
    vim \
    wget \
    php-cli \
    && rm -rf /var/lib/apt/lists/

CMD ["/bin/bash"]

                                [ Read 10 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location  M-U Undo      M-A Set Mark  M-] To Bracket
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy      ^Q Where Was
```

Run WordPress and MySQL Containers Individually on the Same EC2 Instance

```
rom user root. Shutting down mysqld (Version: 8.0.46).
2026-10-14T17:41:25.944974Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Sh
utdown complete (mysqld 8.0.46)  MySQL Community Server - GPL.
2026-10-14 17:41:26+00:00 [Note] [Entrypoint]: Temporary server stopped

2026-10-14 17:41:26+00:00 [Note] [Entrypoint]: MySQL init process done. Ready fo
r start up.

2026-10-14T17:41:26.642416Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-
host-cache' is deprecated and will be removed in a future release. Please use SE
T GLOBAL host_cache_size=0 instead.
2026-10-14T17:41:26.644414Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (my
sqld 8.0.46) starting as process 1
2026-10-14T17:41:26.651645Z 1 [System] [MY-013576] [InnoDB] InnoDB initializatio
n has started.
2026-10-14T17:41:26.908600Z 1 [System] [MY-013577] [InnoDB] InnoDB initializatio
n has ended.
2026-10-14T17:41:27.145993Z 0 [Warning] [MY-010068] [Server] CA certificate ca.p
em is self signed.
2026-10-14T17:41:27.146040Z 0 [System] [MY-013602] [Server] Channel mysql_main c
onfigured to support TLS. Encrypted connections are now supported for this chann
el.
2026-10-14T17:41:27.153124Z 0 [Warning] [MY-011810] [Server] Insecure configurat
ion for --pid-file: Location '/var/run/mysqld' in the path is accessible to all
OS users. Consider choosing a different directory.
2026-10-14T17:41:27.184566Z 0 [System] [MY-011323] [Server] X Plugin ready for c
onnections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
2026-10-14T17:41:27.184788Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: re
ady for connections. Version: '8.0.46'  socket: '/var/run/mysqld/mysqld.sock'  p
ort: 3306  MySQL Community Server - GPL.
ubuntu@ip-172-31-12-203:~$ sudo docker logs wordpress-site
WordPress not found in /var/www/html - copying now...
Complete! WordPress has been successfully copied to /var/www/html
No 'wp-config.php' found in /var/www/html, but 'WORDPRESS_...' variables supplie
d; copying 'wp-config-docker.php' (WORDPRESS_DB_HOST WORDPRESS_DB_NAME WORDPRESS
_DB_PASSWORD WORDPRESS_DB_USER)
AH00558: apache2: Could not reliably determine the server's fully qualified doma
in name, using 172.17.0.3. Set the 'ServerName' directive globally to suppress t
his message
AH00558: apache2: Could not reliably determine the server's fully qualified doma
in name, using 172.17.0.3. Set the 'ServerName' directive globally to suppress t
his message
[Wed Oct 14 17:42:35.509167 2026] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Ap
ache/2.4.68 (Debian) PHP/8.3.35 configured -- resuming normal operations
[Wed Oct 14 17:42:35.509223 2026] [core:notice] [pid 1:tid 1] AH00094: Command l
ine: 'apache2 -D FOREGROUND'
```

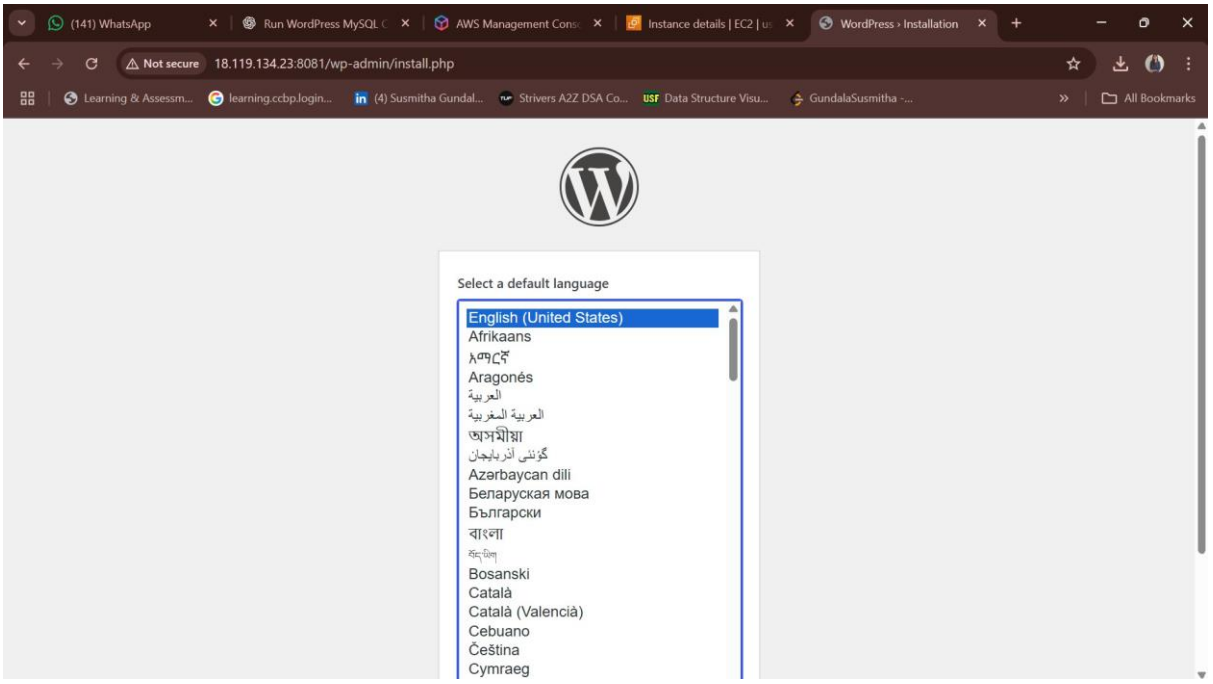
```
docker.io/library/mysql:8.0
ubuntu@ip-172-31-12-203:~$ sudo docker images
                                                    i Info →  U  In Use
IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:8.0          7dcddc01f13b        1.1GB          249MB
wordpress:latest   81a34254acaa       1.12GB          284MB
ubuntu@ip-172-31-12-203:~$ sudo docker run -d \
  --name mysql-db \
  -p 3307:3306 \
  -e MYSQL_ROOT_PASSWORD='RootPass123!' \
  -e MYSQL_DATABASE=wordpress \
  -e MYSQL_USER=wpuser \
  -e MYSQL_PASSWORD='WpPass123!' \
  mysql:8.0
356a8cf19d6b3c9c9d385c16566d8181f96c680dd75810e31d541aa8e12c3f76
ubuntu@ip-172-31-12-203:~$ sudo docker ps
sudo docker logs mysql-db
CONTAINER ID   IMAGE       COMMAND                  CREATED          STATUS
     PORTS                                                  NAMES
356a8cf19d6b   mysql:8.0   "docker-entrypoint.s…"   14 seconds ago   Up 13 secon
ds   33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp   mysql-db
2026-10-14 17:41:13+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Serve
r 8.0.46-1.el9 started.
2026-10-14 17:41:13+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysq
l'
2026-10-14 17:41:13+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Serve
r 8.0.46-1.el9 started.
2026-10-14 17:41:13+00:00 [Note] [Entrypoint]: Initializing database files
2026-10-14T17:41:13.612806Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-
host-cache' is deprecated and will be removed in a future release. Please use SE
```

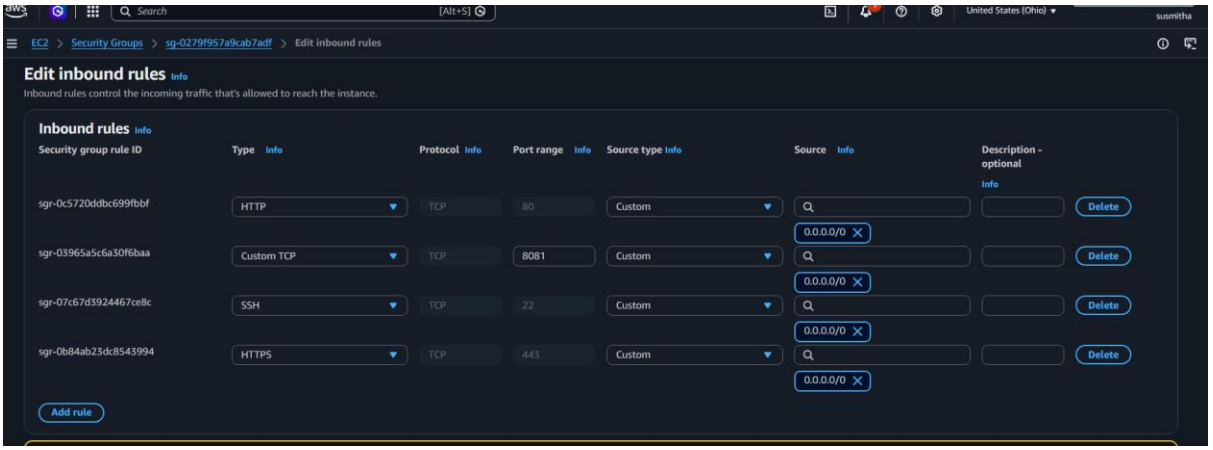
```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-12-203:~$ sudo apt update
Hit:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [12
6 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [
126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packag
es [15.0 MB]
Get:6 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-
en [5982 kB]
Get:7 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Compon
ents [3871 kB]
Get:8 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f
Metadata [301 kB]
Get:9 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Pack
ages [269 kB]
Get:10 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translati
on-en [118 kB]
Get:11 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Com
ponents [35.0 kB]
Get:12 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n
-f Metadata [8328 B]
Get:13 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 P
ackages [1397 kB]
Get:14 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main Transla
tion-en [308 kB]
Get:15 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 C
omponents [181 kB]
Get:16 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c
-n-f Metadata [17.7 kB]
Get:17 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd
64 Packages [1702 kB]
```


(141) WhatsApp
Run WordPress MySQL
AWS Management Cons
Instance details | EC2 | us
WordPress › Installation
Not secure
18.119.134.23:8081/wp-admin/install.php
Learning & Assessm...
learning.ccbp.login...
(4) Susmitha Gundal...
Strivers A2Z DSA Co...
Data Structure Visu...
GundalaSusmitha -...
All Bookmarks
Select a default language
English (United States)
Afrikaans
አማርኛ
Aragonés
العربية
العربية المغربية
অসমীয়া
گؤنئی آذربایجان
Azərbaycan dili
Беларуская мова
Български
বাংলা
Bosanski
Català
Català (Valencià)
Cebuano
Čeština
Cymraeg


Search
[Alt+S]
United States (Ohio)
susmitha
EC2 > Security Groups > sg-0279f957a9cab7adf > Edit inbound rules
Edit inbound rules Info
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules Info
Security group rule ID
Type Info
Protocol Info
Port range Info
Source type Info
Source Info
Description - optional Info
sgr-0c5720ddbc699fbbf
HTTP
TCP
80
Custom
0.0.0.0/0
Delete
sgr-03965a5c6a30f6baa
Custom TCP
TCP
8081
Custom
0.0.0.0/0
Delete
sgr-07c67d3924467ce8c
SSH
TCP
22
Custom
0.0.0.0/0
Delete
sgr-0b84ab23dc8543994
HTTPS
TCP
443
Custom
0.0.0.0/0
Delete
Add rule